# Alessio **Novel**

Artificial Intelligence Student

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## Education

### Università degli Studi di Trento — *Trento, Italy*

M.Sc. in Artificial Intelligence Systems (Code: LM-32) – Neuroscience Track — *Sep 2024 – Present*

- **Current GPA**: 28.8/30
- **Key Courses**: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Foundation of Cognitive Psychology and Neuroscience, Neuroimaging

### Università degli Studi di Trento — *Trento, Italy*

B.Sc. in Computer Science (Code: L-31) — *Sep 2021 – Jul 2024*

- **Graduation Grade**: 110/110 with honors
- **Key Courses**: Algorithms and Data Structures, Databases, Networking, Computer and Network Security, Machine Learning, Human-Computer Interaction, Information Systems

### Istituto Tecnico Statale A. Volta — *Trieste, Italy*

Secondary School Diploma in computer science (Code: D040) — *Sep 2016 – Jun 2021*

- **Graduation Grade**: 100/100

## Work Experience

### CAIAN (Centre for AI and Neuroscience) - University of Bonn — *Bonn, Germany*

Computational Neuroscience & AI Research Intern (Erasmus+ Internship) — *Aug 2026 – Present*

- Conducting research on **representational alignment** within a global multi-lab collaboration collecting the largest single-trial fMRI dataset on **fear conditioning**, supervised by Prof. Dr. Dr. Dominik Bach and Dr. Jurjen Heij.
- Preprocessing fMRI brain imaging data and building geometric analysis pipelines to evaluate human neural representations of fear responses.
- Designing and training brain-inspired Recurrent Neural Networks with Reinforcement Learning to model fear conditioning learning and compare artificial vs. biological neural representations.

## Projects

### Few-Shot CLIP Adaptation (Flowers102 dataset)

GitHub Repository

- Developed and compared new few-shot learning strategies (based on CoOp) to adapt the CLIP vision-language model for fine-grained classification on Oxford Flowers102 dataset.
- The goal was to improve accuracy using only 10 samples for half of the classes, while not having samples for the other half. Improved Base Accuracy by **11.49%** (from 66.92% to 78.41%) and increased the overall Harmonic Mean by **5.75%** compared to the zero-shot baseline, achieving the objective of improving base class accuracy while minimizing degradation on novel class performance.

## Skills

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|---|---|
| **Programming** | Python, C++, Java, R |
| **Back-end** | PHP, REST API, MySQL, PostgreSQL, MongoDB |
| **Front-end** | HTML5, CSS3, JavaScript |
| **DevOps** | Git, Conda, Docker, CMake |
| **Languages** | Italian (native), English |

## Extracurricular Activities

### M2L Summer School — *Milan, Italy*

Participant — *Sep 2024*

- Selected among 519 applicants (34% acceptance rate) for an immersive Machine Learning summer school. Engaged with industry experts on advanced AI topics through intensive technical lectures, workshops, and networking sessions.

### Olympic Winter Games Milano Cortina 2026 — *Tesero, Italy*

Technology Volunteer — *Jul 2025 – Feb 2026*

- After specialized training, I managed IT setup and connectivity for media and staff at the Cross-Country Skiing venue, overseeing device distribution and network support.